

Lab07_Mazo

Gustavo Mazo Vieira

1)

```
library(DBI)
library(RSQLite)

conn <- dbConnect(SQLite(), "uwmadison.sqlite3")
```

2)

```
dbListTables(conn)
```

```
[1] "course_offerings"      "courses"               "grade_distributions"
[4] "instructors"           "rooms"                  "schedules"
[7] "sections"             "subject_memberships"    "subjects"
[10] "teachings"
```

```
professores_stat <- dbGetQuery(conn, "
  SELECT DISTINCT instructors.id,instructors.name FROM subjects
  INNER JOIN subject_memberships ON subjects.code = subject_memberships.subject_code
  INNER JOIN course_offerings ON subject_memberships.course_offering_uuid = course_offerings
  INNER JOIN sections ON course_offerings.uuid = sections.course_offering_uuid
  INNER JOIN teachings ON sections.uuid = teachings.section_uuid
  INNER JOIN instructors ON teachings.instructor_id = instructors.id
  WHERE subjects.abbreviation = 'STAT'
  ORDER BY instructors.name
")

nrow(professores_stat)
```

```
[1] 379
```

3)

```
gpa_stat <- dbGetQuery(conn, "
SELECT
  course_offerings.uuid AS course_offering_uuid,
  courses.number AS disciplina,
  courses.name AS nome_disciplina,
  grade_distributions.section_number,

  (
    4.0 * grade_distributions.a_count +
    3.5 * grade_distributions.ab_count +
    3.0 * grade_distributions.b_count +
    2.5 * grade_distributions.bc_count +
    2.0 * grade_distributions.c_count +
    1.0 * grade_distributions.d_count +
    0.0 * grade_distributions.f_count
  )
  /
  NULLIF(
    grade_distributions.a_count +
    grade_distributions.ab_count +
    grade_distributions.b_count +
    grade_distributions.bc_count +
    grade_distributions.c_count +
    grade_distributions.d_count +
    grade_distributions.f_count,
    0
  ) AS gpa

FROM subjects

INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code

INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid = course_offerings.uuid

INNER JOIN courses
  ON course_offerings.course_uuid = courses.uuid

INNER JOIN grade_distributions
  ON course_offerings.uuid = grade_distributions.course_offering_uuid
```

```
WHERE subjects.abbreviation = 'STAT'
")
```

```
professor_dificil <- dbGetQuery(conn, "
SELECT
  instructors.name AS professor,

  AVG(
    (
      4.0 * grade_distributions.a_count +
      3.5 * grade_distributions.ab_count +
      3.0 * grade_distributions.b_count +
      2.5 * grade_distributions.bc_count +
      2.0 * grade_distributions.c_count +
      1.0 * grade_distributions.d_count +
      0.0 * grade_distributions.f_count
    )
    /
    NULLIF(
      grade_distributions.a_count +
      grade_distributions.ab_count +
      grade_distributions.b_count +
      grade_distributions.bc_count +
      grade_distributions.c_count +
      grade_distributions.d_count +
      grade_distributions.f_count,
      0
    )
  ) AS gpa_medio

FROM subjects

INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code

INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid = course_offerings.uuid

INNER JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid

INNER JOIN grade_distributions
```

```

    ON sections.course_offering_uuid = grade_distributions.course_offering_uuid
    AND sections.number = grade_distributions.section_number

INNER JOIN teachings
    ON sections.uuid = teachings.section_uuid

INNER JOIN instructors
    ON teachings.instructor_id = instructors.id

WHERE subjects.abbreviation = 'STAT'
    AND instructors.name IS NOT NULL

GROUP BY instructors.id, instructors.name

HAVING SUM(
    grade_distributions.a_count +
    grade_distributions.ab_count +
    grade_distributions.b_count +
    grade_distributions.bc_count +
    grade_distributions.c_count +
    grade_distributions.d_count +
    grade_distributions.f_count
) > 0

ORDER BY gpa_medio ASC

LIMIT 1
")
professor_dificil

```

```

    professor gpa_medio
1 JAMES D KUELBS    2.6375

```

```

professor_facil <- dbGetQuery(conn, "
SELECT
    instructors.name AS professor,

    AVG(
        (
            4.0 * grade_distributions.a_count +

```

```

        3.5 * grade_distributions.ab_count +
        3.0 * grade_distributions.b_count +
        2.5 * grade_distributions.bc_count +
        2.0 * grade_distributions.c_count +
        1.0 * grade_distributions.d_count +
        0.0 * grade_distributions.f_count
    )
    /
    NULLIF(
        grade_distributions.a_count +
        grade_distributions.ab_count +
        grade_distributions.b_count +
        grade_distributions.bc_count +
        grade_distributions.c_count +
        grade_distributions.d_count +
        grade_distributions.f_count,
        0
    )
) AS gpa_medio

FROM subjects

INNER JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code

INNER JOIN course_offerings
    ON subject_memberships.course_offering_uuid = course_offerings.uuid

INNER JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid

INNER JOIN grade_distributions
    ON sections.course_offering_uuid = grade_distributions.course_offering_uuid
    AND sections.number = grade_distributions.section_number

INNER JOIN teachings
    ON sections.uuid = teachings.section_uuid

INNER JOIN instructors
    ON teachings.instructor_id = instructors.id

WHERE subjects.abbreviation = 'STAT'

```

```

GROUP BY instructors.id, instructors.name

ORDER BY gpa_medio DESC

LIMIT 1
")
professor_facil

```

```

                professor gpa_medio
1 SRIKANTHMADHAVAN ARAVAMUTHAN          4

```

```

disciplina_dificil <- dbGetQuery(conn, "
SELECT
  courses.number AS disciplina,
  courses.name AS nome_disciplina,

  AVG(
    (
      4.0 * grade_distributions.a_count +
      3.5 * grade_distributions.ab_count +
      3.0 * grade_distributions.b_count +
      2.5 * grade_distributions.bc_count +
      2.0 * grade_distributions.c_count +
      1.0 * grade_distributions.d_count +
      0.0 * grade_distributions.f_count
    )
    /
    NULLIF(
      grade_distributions.a_count +
      grade_distributions.ab_count +
      grade_distributions.b_count +
      grade_distributions.bc_count +
      grade_distributions.c_count +
      grade_distributions.d_count +
      grade_distributions.f_count,
      0
    )
  ) AS gpa_medio

FROM subjects

```

```

INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code

INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid = course_offerings.uuid

INNER JOIN courses
  ON course_offerings.course_uuid = courses.uuid

INNER JOIN grade_distributions
  ON course_offerings.uuid = grade_distributions.course_offering_uuid

WHERE subjects.abbreviation = 'STAT'
  AND courses.name IS NOT NULL

GROUP BY courses.uuid, courses.number, courses.name

HAVING SUM(
  grade_distributions.a_count +
  grade_distributions.ab_count +
  grade_distributions.b_count +
  grade_distributions.bc_count +
  grade_distributions.c_count +
  grade_distributions.d_count +
  grade_distributions.f_count
) > 0

ORDER BY gpa_medio ASC

LIMIT 1
")
disciplina_dificil

```

```

disciplina          nome_disciplina gpa_medio
1      431 Introduction to the Theory of Probability  2.891568

```

```

disciplina_facil <- dbGetQuery(conn, "
SELECT
  courses.number AS disciplina,
  courses.name AS nome_disciplina,

```

```

AVG(
  (
    4.0 * grade_distributions.a_count +
    3.5 * grade_distributions.ab_count +
    3.0 * grade_distributions.b_count +
    2.5 * grade_distributions.bc_count +
    2.0 * grade_distributions.c_count +
    1.0 * grade_distributions.d_count +
    0.0 * grade_distributions.f_count
  )
  /
  NULLIF(
    grade_distributions.a_count +
    grade_distributions.ab_count +
    grade_distributions.b_count +
    grade_distributions.bc_count +
    grade_distributions.c_count +
    grade_distributions.d_count +
    grade_distributions.f_count,
    0
  )
) AS gpa_medio

FROM subjects

INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code

INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid = course_offerings.uuid

INNER JOIN courses
  ON course_offerings.course_uuid = courses.uuid

INNER JOIN grade_distributions
  ON course_offerings.uuid = grade_distributions.course_offering_uuid

WHERE subjects.abbreviation = 'STAT'

GROUP BY courses.uuid, courses.number, courses.name

ORDER BY gpa_medio DESC

```

```
LIMIT 1
```

```
")
```

```
disciplina_facil
```

```
disciplina                                nome_disciplina gpa_medio
1      841 Nonparametric Statistics and Machine Learning Methods      4
```

```
4)
```

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
```

```
Sys.time()
```

```
[1] "2026-09-22 15:04:48 -03"
```